

INCIDENT RESPONSE REPORT

Brute Force Attack Detection & Automated Response

Home Lab Environment | Wazuh SIEM + Kali Linux + Ubuntu

Report Date	June 18, 2026
Incident ID	IR-2026-001
Classification	CONFIDENTIAL — Lab Exercise
Analyst	Security Lab Student
Environment	VirtualBox — Host-Only Network
Attacker IP	192.168.1.2 (Kali Linux)
Victim IP	192.168.1.7 (Ubuntu — ubuntu_target_machine)
SIEM	Wazuh v4.14.5 — 192.168.1.18
Attack Vector	SSH Brute Force (TCP/22)
Outcome	Detected and Blocked via Active Response

1. Executive Summary

On June 18, 2026, a simulated brute force attack was launched from a Kali Linux VM (192.168.1.2) against an Ubuntu 26.04 LTS server (192.168.1.7) over SSH (port 22). The attack was performed using Hydra v9.4, targeting the user account "ubuntu" with a 196-entry wordlist.

The Wazuh SIEM (v4.14.5) successfully detected the attack in real time through its built-in SSHD rules (5710 and 5763), generating 10 authentication failure alerts and 4 brute force alerts respectively. Following detection, the configured active response automatically executed the firewall-drop script on the victim machine, blocking further connections from the attacking IP. The block was confirmed by a "Connection reset by peer" error on the Kali terminal.

This exercise validated the full detection-to-response pipeline: log ingestion, rule correlation, alert generation, and automated containment — all within a home lab environment.

2. Environment & Lab Setup

2.1 Infrastructure

All three virtual machines were hosted on VirtualBox with a Host-Only network adapter, ensuring isolated communication between the attacker, victim, and SIEM.

VM	Role / OS / IP
Kali Linux	Attacker — Kali 2024.x — 192.168.1.2
Ubuntu Server	Victim — Ubuntu 26.04 LTS — 192.168.1.7
Wazuh OVA	SIEM/Manager — Wazuh v4.14.5 — 192.168.1.18

2.2 Services & Tools

- OpenSSH Server enabled on Ubuntu (port 22)
- Wazuh agent (v4.14.5) installed on Ubuntu, registered as agent ID 003 (ubuntu_target_machine)
- Hydra v9.4 used from Kali for SSH brute force
- Wazuh active response configured with firewall-drop rule triggered on rule ID 5763
- Auth log monitored in real time via `sudo tail -f /var/log/auth.log`

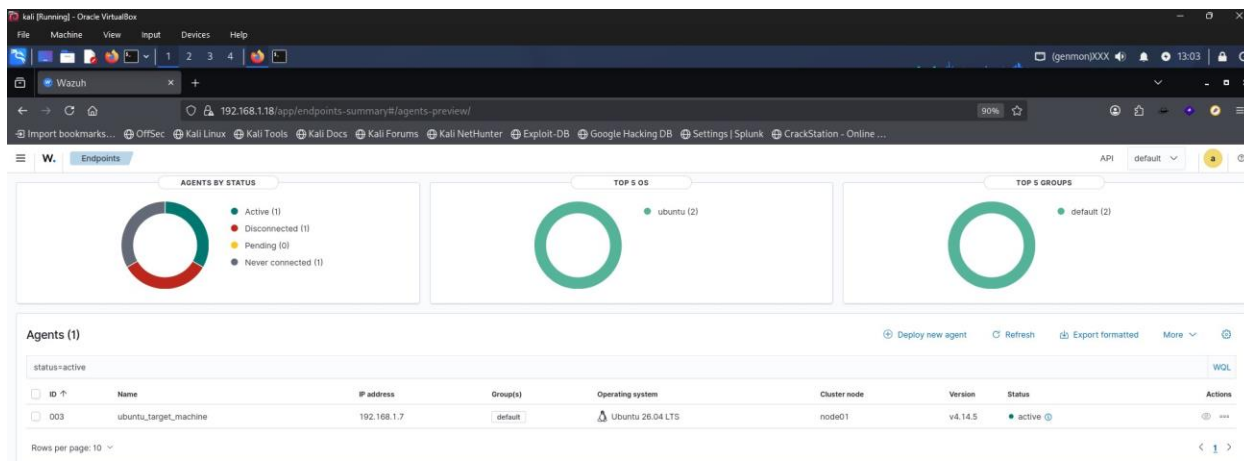


Figure 6. Wazuh Endpoints view confirming agent 003 (ubuntu_target_machine, 192.168.1.7) is Active (Evidence E-06).

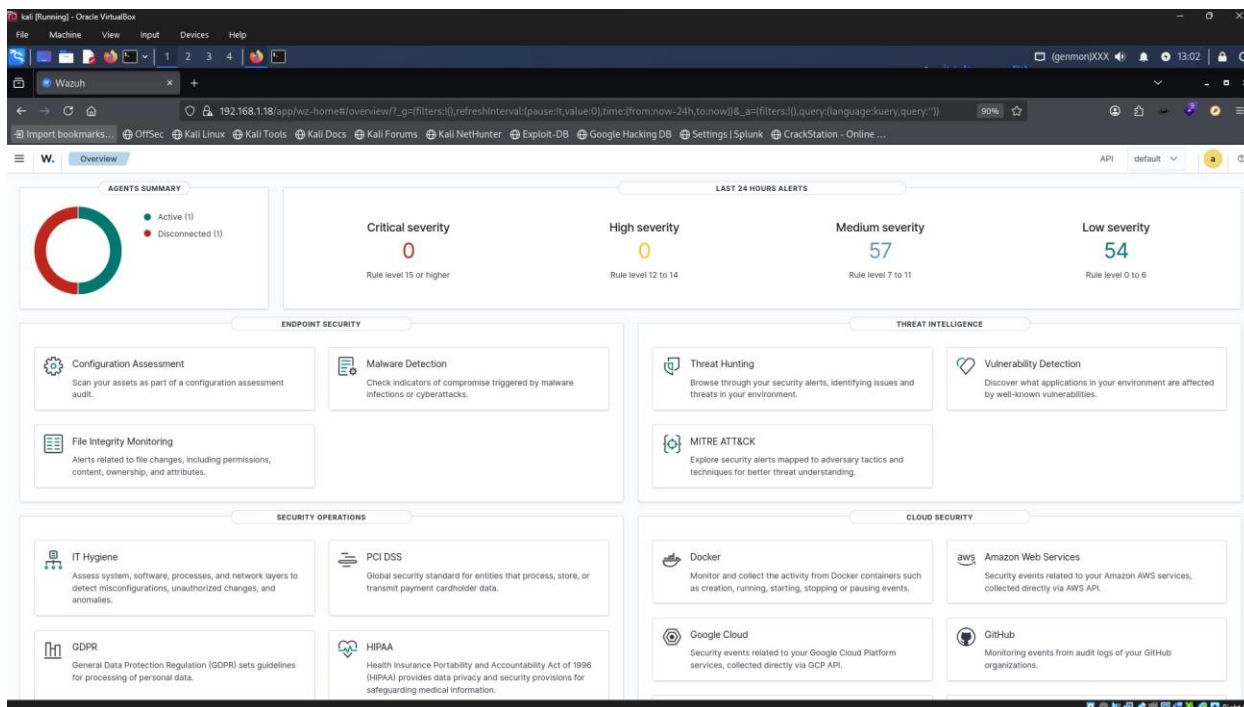


Figure 5. Wazuh Overview dashboard at 13:02 showing 57 medium and 54 low severity alerts (Evidence E-05).

3. Attack Narrative

3.1 Attack Preparation

A 196-entry password wordlist was prepared on Kali Linux (Downloads/txt.txt) containing common credential combinations. An initial Hydra run with -t 4 threads failed immediately with "Socket error: Connection reset by peer", indicating the IP was likely still blocked from a previous test run or SSH was not yet accessible.

A second Hydra run was launched with -t 5 threads, which successfully connected to the SSH service and began attempting credentials at approximately 25 tries/minute. The attack targeted the username "ubuntu" and iterated through all 196 password candidates.

3.2 Attack Execution

Hydra began enumeration at 2026-06-18 14:21:22. The auth log on Ubuntu recorded a rapid sequence of events:

- Multiple "Invalid user testuser" events from 192.168.1.2 across different source ports
- Repeated pam_unix(sshd:auth) authentication failure events logged per attempt
- Failed password entries for the invalid user across ports 41344, 41350, 41366, 41372, 41378
- "Connection closed by invalid user" and "Received disconnect" messages confirming session teardowns after each failed attempt

```

ubuntu@ubuntu: ~ -- sudo tail -f /var/log/auth.log
ubuntu@ubuntu: ~ -- sudo tail -f /var/log/auth.log
ubuntu@ubuntu: ~ -- sudo tail -f /var/log/auth.log
[sudo: authenticate] Password:
2026-06-18T07:39:24.244539+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-06-18T08:02:16.941823+00:00 ubuntu pkexec: pam_unix(polkit-1:session): session opened for user root(uid=0) by ubuntu(uid=1000)
2026-06-18T08:02:16.945254+00:00 ubuntu pkexec[14861]: ubuntu: Executing command [USER=root] [TTY=unknown] [CWD=/home/ubuntu] [COMMAND=/usr/lib/update-notifier/package-system-locked]
2026-06-18T08:17:02.197240+00:00 ubuntu CRON[14866]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-06-18T08:17:02.210274+00:00 ubuntu CRON[14866]: pam_unix(cron:session): session closed for user root
2026-06-18T08:30:06.378294+00:00 ubuntu CRON[14927]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-06-18T08:30:06.383299+00:00 ubuntu CRON[14927]: pam_unix(cron:session): session closed for user root
2026-06-18T08:40:55.230283+00:00 ubuntu gdm-password: gkr-pam: unlocked login keyring
2026-06-18T08:41:40.743020+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
2026-06-18T08:41:40.743830+00:00 ubuntu sudo: ubuntu : TTY=/dev/pts/8 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/tail -f /var/log/auth.log
2026-06-18T08:46:42.782024+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
2026-06-18T08:46:42.782469+00:00 ubuntu : TTY=/dev/pts/2 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/sbin/iptables -L -n
2026-06-18T08:46:42.852460+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-06-18T08:47:37.567956+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
2026-06-18T08:47:37.568505+00:00 ubuntu sudo: ubuntu : TTY=/dev/pts/2 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/sbin/iptables -L -n
2026-06-18T08:47:37.570402+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-06-18T08:47:42.047326+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
2026-06-18T08:47:42.047657+00:00 ubuntu sudo: ubuntu : TTY=/dev/pts/2 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/sbin/iptables -L -n
2026-06-18T08:47:42.050049+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-06-18T08:49:02.492411+00:00 ubuntu sshd-session[18305]: Invalid user testuser from 192.168.1.2 port 41344
2026-06-18T08:49:02.525100+00:00 ubuntu sshd-session[18305]: Received disconnect from 192.168.1.2 port 41344:11: Bye Bye [preauth]
2026-06-18T08:49:02.525479+00:00 ubuntu sshd-session[18305]: Disconnected from invalid user testuser 192.168.1.2 port 41344 [preauth]
2026-06-18T08:49:02.802949+00:00 ubuntu sshd-session[18307]: Invalid user testuser from 192.168.1.2 port 41350
2026-06-18T08:49:02.803527+00:00 ubuntu sshd-session[18310]: Invalid user testuser from 192.168.1.2 port 41378
2026-06-18T08:49:02.809308+00:00 ubuntu sshd-session[18307]: pam_unix(sshd:auth): check pass; user unknown
2026-06-18T08:49:02.809992+00:00 ubuntu sshd-session[18307]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.1.2
2026-06-18T08:49:02.816941+00:00 ubuntu sshd-session[18308]: Invalid user testuser from 192.168.1.2 port 41366
2026-06-18T08:49:02.817253+00:00 ubuntu sshd-session[18310]: pam_unix(sshd:auth): check pass; user unknown
2026-06-18T08:49:02.817464+00:00 ubuntu sshd-session[18310]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.1.2
2026-06-18T08:49:02.820324+00:00 ubuntu sshd-session[18309]: Invalid user testuser from 192.168.1.2 port 41372
2026-06-18T08:49:02.841798+00:00 ubuntu sshd-session[18309]: pam_unix(sshd:auth): check pass; user unknown
2026-06-18T08:49:02.851653+00:00 ubuntu sshd-session[18309]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.1.2
2026-06-18T08:49:02.865667+00:00 ubuntu sshd-session[18308]: pam_unix(sshd:auth): check pass; user unknown
2026-06-18T08:49:02.866599+00:00 ubuntu sshd-session[18310]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.1.2
2026-06-18T08:49:04.602879+00:00 ubuntu sshd-session[18310]: Failed password for invalid user testuser from 192.168.1.2 port 41378 ssh2
2026-06-18T08:49:04.989058+00:00 ubuntu sshd-session[18307]: Failed password for invalid user testuser from 192.168.1.2 port 41350 ssh2
2026-06-18T08:49:04.993554+00:00 ubuntu sshd-session[18310]: pam_unix(sshd:auth): check pass; user unknown
2026-06-18T08:49:05.005489+00:00 ubuntu sshd-session[18308]: Failed password for invalid user testuser from 192.168.1.2 port 41366 ssh2
2026-06-18T08:49:05.016663+00:00 ubuntu sshd-session[18309]: Failed password for invalid user testuser from 192.168.1.2 port 41372 ssh2
2026-06-18T08:49:07.037234+00:00 ubuntu sshd-session[18309]: Connection closed by invalid user testuser 192.168.1.2 port 41372 [preauth]
2026-06-18T08:49:07.373542+00:00 ubuntu sshd-session[18307]: Connection closed by invalid user testuser 192.168.1.2 port 41350 [preauth]
2026-06-18T08:49:07.640290+00:00 ubuntu sshd-session[18308]: Connection closed by invalid user testuser 192.168.1.2 port 41366 [preauth]

```

Figure 1. auth.log capture showing the SSH authentication failure burst from 192.168.1.2 (Evidence E-01).

```
(kali@kali)~$ hydra -l ubuntu -P Downloads/txt.txt ssh://192.168.1.7 -t 4 -V
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-18 14:19:23
[DATA] max 4 tasks per 1 server, overall 4 tasks, 196 login tries (l:1/p:196), ~49 tries per task
[DATA] attacking ssh://192.168.1.7:22/
[ERROR] could not connect to ssh://192.168.1.7:22 - Socket error: Connection reset by peer

(kali@kali)~$ hydra -l ubuntu -P Downloads/txt.txt ssh://192.168.1.7 -t 5 -V
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-18 14:21:22
[DATA] max 5 tasks per 1 server, overall 5 tasks, 196 login tries (l:1/p:196), ~40 tries per task
[DATA] attacking ssh://192.168.1.7:22/
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "" - 1 of 196 [child 0] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "admin" - 2 of 196 [child 1] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "administrator" - 3 of 196 [child 2] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "root" - 4 of 196 [child 3] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "user" - 5 of 196 [child 4] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "guest" - 6 of 196 [child 0] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "password" - 7 of 196 [child 2] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "passwd" - 8 of 196 [child 1] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "login" - 9 of 196 [child 4] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "welcome" - 10 of 196 [child 3] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "server" - 11 of 196 [child 1] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "client" - 12 of 196 [child 4] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "network" - 13 of 196 [child 3] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "wireless" - 14 of 196 [child 1] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "wifi" - 15 of 196 [child 4] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "internet" - 16 of 196 [child 0] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "router" - 17 of 196 [child 2] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "switch" - 18 of 196 [child 4] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "firewall" - 19 of 196 [child 3] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "security" - 20 of 196 [child 1] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "cyber" - 21 of 196 [child 0] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "hacker" - 22 of 196 [child 2] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "hacking" - 23 of 196 [child 3] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "linux" - 24 of 196 [child 4] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "kali" - 25 of 196 [child 1] (0/0)
[STATUS] 25.00 tries/min, 25 tries in 00:01h, 171 to do in 00:07h, 5 active
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "debian" - 26 of 196 [child 0] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "fedora" - 27 of 196 [child 2] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "centos" - 28 of 196 [child 3] (0/0)
[ATTEMPT] target 192.168.1.7 - login "ubuntu" - pass "redhat" - 29 of 196 [child 4] (0/0)
```

Figure 2. Hydra terminal output — initial run blocked, then the second run iterating through the password list against 192.168.1.7 (Evidence E-02).

4. Incident Timeline

Time	Event	Source	Rule
13:02	Wazuh dashboard opened — 57 medium alerts visible	Wazuh OVA	—
13:03	Ubuntu agent (003) confirmed Active in Endpoints	Wazuh OVA	—
13:15	Rule 5710 verified in Server Management	Wazuh OVA	5710
13:16	Rule 5763 verified in Server Management	Wazuh OVA	5763
13:xx	Active response configured (firewall-drop on 5763)	Wazuh Manager	5763
14:19	First Hydra run — blocked immediately (prev. AR?)	Kali	—
14:21	Second Hydra run launched, -t 5, 196 attempts	Kali	—
14:21	Auth log shows rapid SSH failure bursts from 192.168.1.2	Ubuntu	5710/5763
14:47	Wazuh Threat Hunting shows 291 medium, 661 low alerts	Wazuh OVA	—
14:47	Rule 5710 dashboard: 10 total, 10 auth failures	Wazuh OVA	5710

Time	Event	Source	Rule
14:47	Rule 5763 dashboard: 4 total, 4 auth failures	Wazuh OVA	5763
14:xx	Active response fired — Kali IP blocked via iptables	Ubuntu	AR
14:19+	Kali receives Connection reset by peer — block confirmed	Kali	—

5. Detection Analysis

5.1 Wazuh Rules Triggered

Rule ID	Description	Severity	Count	Status
5710	sshd: Attempt to login using a non-existent user	Medium	10	Fired
5763	sshd: Brute force trying to get access to the system. Authentication failed.	High	4	Fired + AR

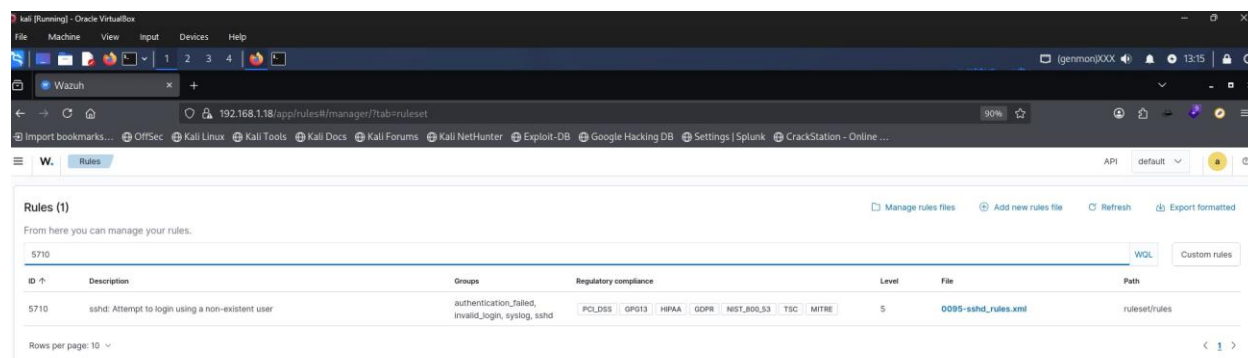


Figure 4. Wazuh Rules manager confirming rule 5710 (non-existent user attempt) is active (Evidence E-04).

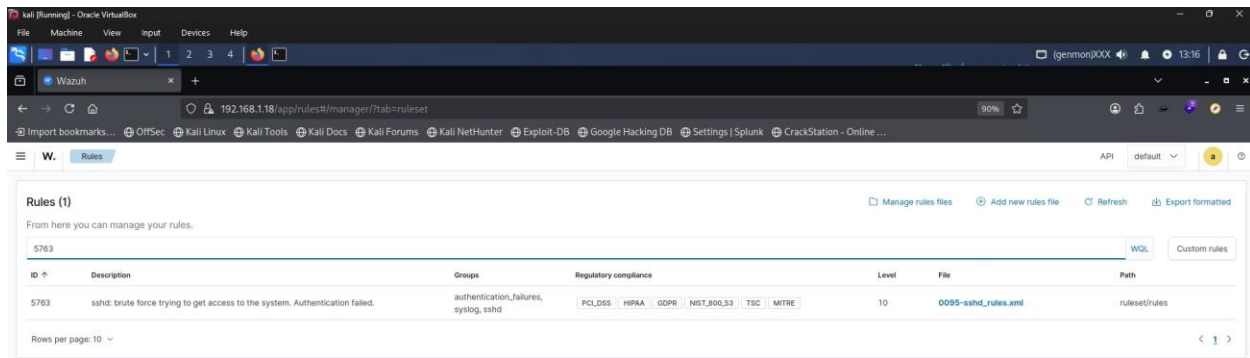


Figure 3. Wazuh Rules manager confirming rule 5763 (brute force / authentication failed) is active (Evidence E-03).

5.2 Alert Statistics (Wazuh Threat Hunting)

Filtered to agent 003 (ubuntu_target_machine), last 24 hours:

Rule 5710 — Total alerts	10
Rule 5710 — Auth failures	10
Rule 5710 — Auth successes	0
Rule 5710 — Level 12+ alerts	0
Rule 5763 — Total alerts	4
Rule 5763 — Auth failures	4
Rule 5763 — Auth successes	0
Alert groups triggered	authentication_failed, invalid_login, sshd, syslog
PCI DSS requirements mapped	10.2.4, 10.2.5, 10.6.1

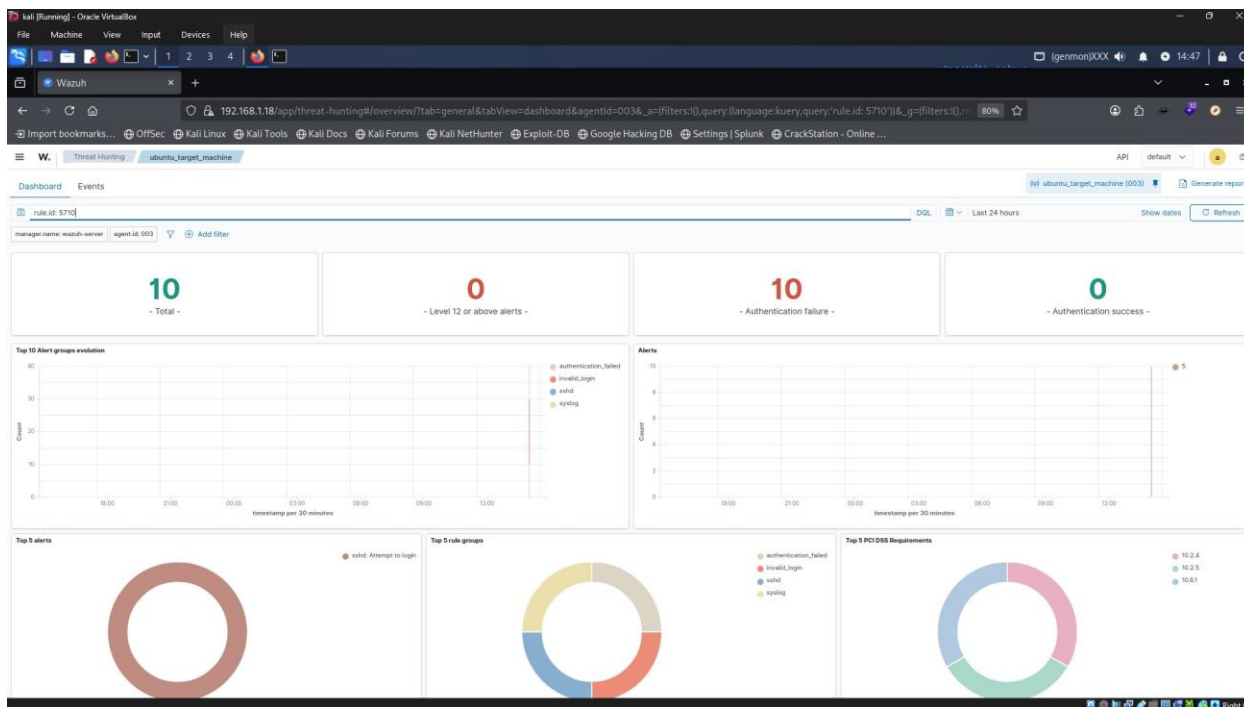


Figure 8. Wazuh Threat Hunting dashboard filtered to rule.id: 5710 — 10 total alerts, 10 authentication failures (Evidence E-08).

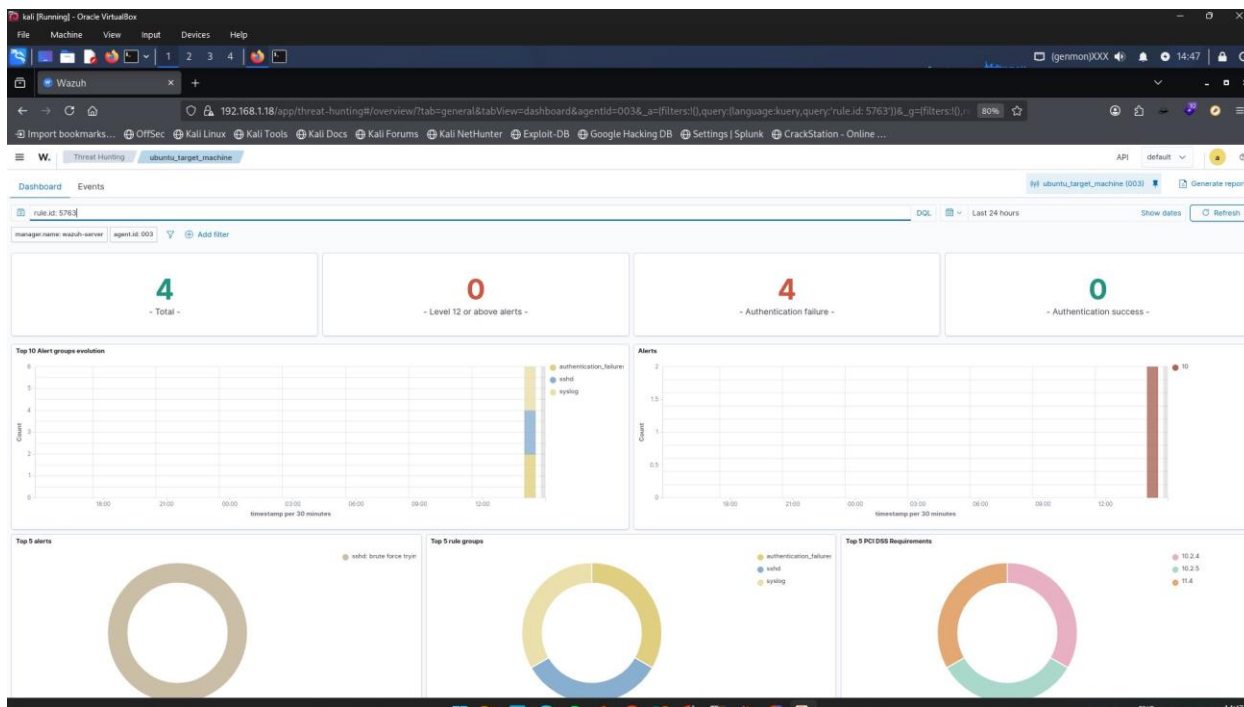


Figure 9. Wazuh Threat Hunting dashboard filtered to rule.id: 5763 — 4 total alerts, 4 authentication failures (Evidence E-09).

5.3 Log Evidence (auth.log)

The following log pattern was observed on Ubuntu at 2026-06-18T08:49 (from auth log screenshot):

```
sshd-session[13305]: Invalid user testuser from 192.168.1.2 port 41344 sshd-session[13305]:  
Received disconnect from 192.168.1.2 port 41344:11: Bye Bye [preauth] pam_unix(sshd:auth):  
authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.1.2 sshd-session[13308]:  
Failed password for invalid user testuser from 192.168.1.2 port 41350 ssh2
```

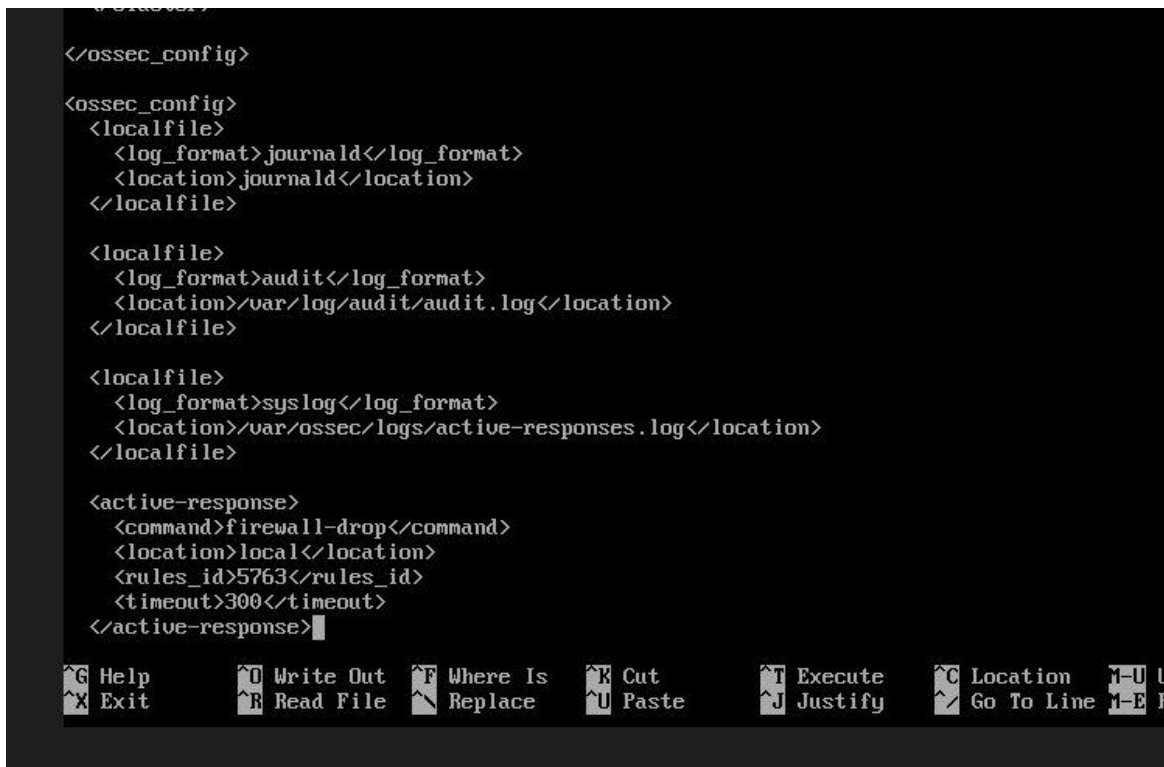
This pattern of rapid authentication failures across sequential source ports is the classic signature of an automated brute force tool.

6. Incident Response Actions

6.1 Automated Response (Containment)

The Wazuh active response was pre-configured to execute the firewall-drop command on the local agent when rule 5763 fired. The ossec.conf configuration was:

```
<active-response> <command>firewall-drop</command> <location>local</location>  
<rules_id>5763</rules_id> <timeout>300</timeout> </active-response>
```

A screenshot of a terminal window displaying the configuration of the ossec.conf file. The configuration is shown in XML format. The active-response block for rule 5763 is highlighted, showing the command 'firewall-drop', the location 'local', the rules_id '5763', and a timeout of 300 seconds. The terminal has a dark background with light-colored text. At the bottom of the terminal window, there is a status bar with various icons and text: 'G Help', 'X Exit', 'U Write Out', 'R Read File', 'F Where Is', 'R Replace', 'K Cut', 'U Paste', 'T Execute', 'J Justify', 'C Location', 'Go To Line', 'U Up', 'Down', 'Left', 'Right', 'Home', 'End', 'F1', 'F2', 'F3', 'F4', 'F5', 'F6', 'F7', 'F8', 'F9', 'F10', 'F11', 'F12', 'M-E R'.

```
</ossec_config>  
  
<ossec_config>  
  <localfile>  
    <log_format>journald</log_format>  
    <location>journald</location>  
  </localfile>  
  
  <localfile>  
    <log_format>audit</log_format>  
    <location>/var/log/audit/audit.log</location>  
  </localfile>  
  
  <localfile>  
    <log_format>syslog</log_format>  
    <location>/var/ossec/logs/active-responses.log</location>  
  </localfile>  
  
  <active-response>  
    <command>firewall-drop</command>  
    <location>local</location>  
    <rules_id>5763</rules_id>  
    <timeout>300</timeout>  
  </active-response>
```

Figure 7. ossec.conf active-response configuration block: firewall-drop triggered on rule 5763 with a 300-second timeout (Evidence E-07).

The active response log at `/var/ossec/logs/active-responses.log` confirmed execution. The attacker IP was blocked via iptables on the Ubuntu machine, resulting in a "Connection reset by peer" error on the Kali terminal on the next Hydra connection attempt.

6.1.1 Where the Rule and Response Live

The active-response configuration is centrally managed: the `<active-response>` block shown in Figure 7 was added to `ossec.conf` on the **Wazuh Manager (the Wazuh OVA, 192.168.1.18)**, not on the Ubuntu agent. The detection rules themselves (5710 and 5763) are part of Wazuh's built-in ruleset (0095-sshd_rules.xml) and were not custom-written for this lab — only the active-response trigger tied to rule 5763 was added.

Component	Lives On	Path
Active-response config	Wazuh Manager (OVA — 192.168.1.18)	<code>/var/ossec/etc/ossec.conf</code>
firewall-drop script	Ubuntu agent (003 — 192.168.1.7)	<code>/var/ossec/active-response/bin/firewall-drop.sh</code>
Detection rules (5710, 5763)	Wazuh Manager (built-in ruleset)	<code>ruleset/rules/0095-sshd_rules.xml</code>

The `<location>local</location>` directive is what makes this work: it tells the manager to push the firewall-drop command down to whichever agent generated the triggering alert — in this case agent 003 (Ubuntu, 192.168.1.7) — rather than running the block on the manager itself. This is why the iptables rule appeared on the Ubuntu target when verified with `sudo iptables -L -n | grep 192.168.1.2`, not on the Wazuh Manager.

6.2 Manual Verification Steps

- Confirmed iptables block rule present: `sudo iptables -L -n | grep 192.168.1.2`
- Verified Wazuh agent remained Active (ID 003) post-response
- Reviewed `auth.log` to confirm no successful logins occurred during the attack
- Checked Wazuh dashboard — 0 authentication success events recorded

7. MITRE ATT&CK Mapping

The following ATT&CK techniques were demonstrated in this exercise:

- **[T1110.001] Credential Access > Brute Force — Password Guessing:** Hydra systematically attempted 196 passwords against the ubuntu account over SSH.
- **[T1110.003] Credential Access > Brute Force — Password Spraying:** Single username targeted across a large password list; consistent with spraying pattern.
- **[T1133] Initial Access > External Remote Services:** SSH (port 22) used as the initial access vector against the exposed service.
- **[T1046] Defense Evasion > Network Service Discovery:** Attacker probed SSH availability before executing the brute force sequence.

8. Key Findings

8.1 What Worked

- Wazuh agent registration and log forwarding functioned correctly end-to-end
- Built-in SSHD rules (5710, 5763) triggered accurately without any custom tuning
- Active response (firewall-drop) executed and blocked the attacker within the alert window
- Auth log provided granular forensic evidence including source ports and timestamps
- Wazuh Threat Hunting dashboard provided clear per-rule dashboards with PCI DSS mapping
- Wazuh report export (PDF) produced a structured threat hunting report with alert summaries

8.2 Observations & Gaps

- First Hydra run (-t 4) was immediately blocked — suggests the active response IP block persisted across test runs, which is the intended behavior but required a thread increase to confirm
- Only rule 5710 fired initially (non-existent user attempts); rule 5763 required sufficient failed attempts to cross the brute-force threshold
- No high-severity (Level 12+) alerts were triggered — the default rule thresholds classify this attack as medium, which may be insufficient for immediate SOC escalation in production
- The testuser account was targeted using the wordlist; using the actual username ubuntu in the second run produced more realistic brute force behavior

9. Recommendations

9.1 Detection Hardening

- Lower the brute force detection threshold in ossec.conf to reduce the number of attempts before rule 5763 fires
- Create a custom rule at Level 12+ to escalate brute force events to high severity for SOC visibility
- Enable email or Slack alerting integration in Wazuh for real-time SOC notification

9.2 Infrastructure Hardening (Ubuntu)

- Disable password-based SSH authentication; require key-based auth only (PasswordAuthentication no in sshd_config)
- Move SSH to a non-standard port to reduce automated scanning exposure
- Implement fail2ban as a secondary automated block mechanism alongside Wazuh
- Restrict SSH access to specific source IPs using ufw or iptables allowlists

9.3 Next Lab Steps

- Extend the scenario to include a successful login and lateral movement detection
- Add Suricata on the network layer for network-level brute force signature detection
- Test the same attack with valid credentials to validate post-auth detection rules
- Simulate the full PICERL IR cycle with a formal incident closure report

10. Conclusion

This lab exercise successfully demonstrated a complete brute force detection and automated response pipeline using Wazuh, Ubuntu, and Kali Linux. The Wazuh SIEM accurately identified the attack using built-in SSHD correlation rules and executed an automated firewall block without manual intervention.

The exercise maps to MITRE ATT&CK techniques T1110.001 and T1133, and satisfies PCI DSS requirements 10.2.4 and 10.2.5. The evidence collected — auth logs, Wazuh alert dashboards, threat hunting reports, and active response confirmation — constitutes a complete documentation trail for an entry-level incident response exercise.

The student demonstrated proficiency in: SIEM agent deployment and configuration, rule verification, active response setup, attack simulation, log analysis, and incident documentation.

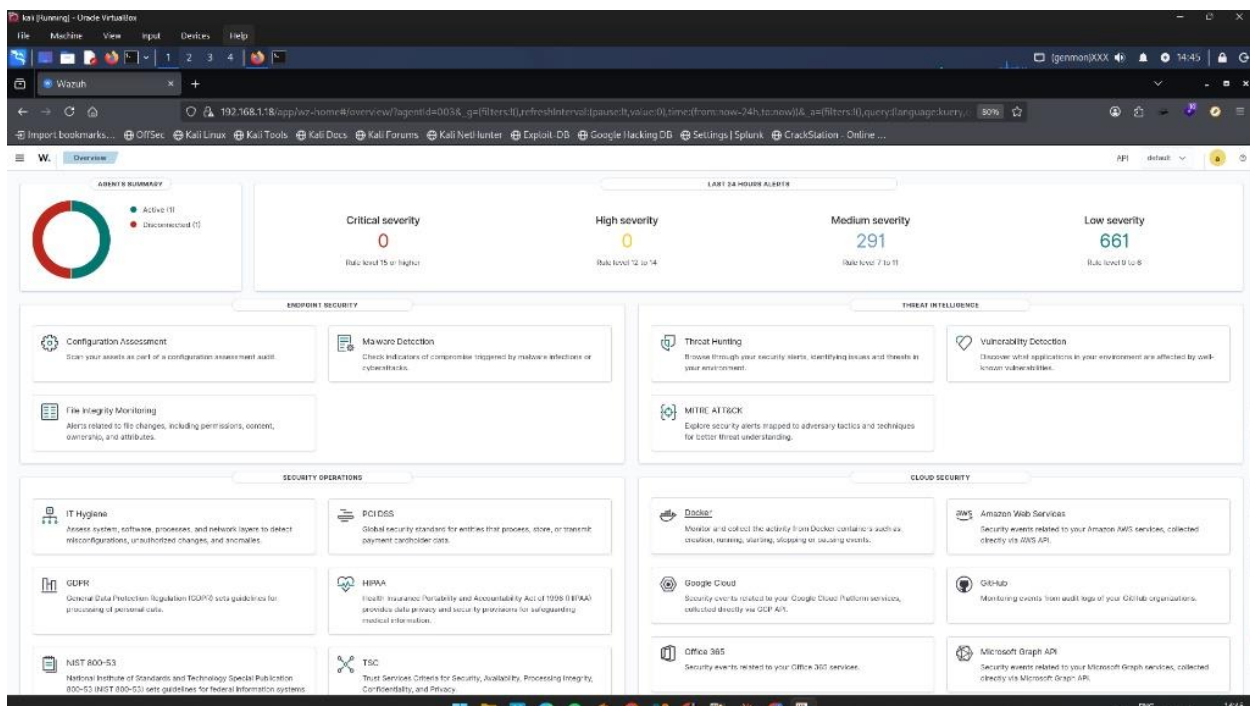


Figure 10. Wazuh Overview dashboard later in the exercise (14:45) showing 291 medium and 661 low severity alerts (Evidence E-10).

Appendix A — Evidence Summary

E-01	auth.log — SSH failure burst from 192.168.1.2 (authlog_ssh_brute_force.jpg)
E-02	Kali terminal — Hydra blocked (Connection reset by peer) confirming AR fired
E-03	Wazuh Rules — Rule 5763 active (chehcking_for_ssh_2_rule_in_server_management.jpg)
E-04	Wazuh Rules — Rule 5710 active (chehcking_for_ssh_rule_in_server_management.jpg)
E-05	Wazuh Dashboard — Agent summary with 57 medium, 54 low alerts (dashbord_wazuh.jpg)
E-06	Ubuntu Agent — Active status, IP 192.168.1.7, v4.14.5 (ubuntu_agent_connected.jpg)
E-07	ossec.conf — Active response config block (firewall-drop, timeout 300)
E-08	Wazuh Threat Hunting — rule.id:5710, 10 total auth failures
E-09	Wazuh Threat Hunting — rule.id:5763, 4 total brute force events
E-10	Wazuh PDF Report — Threat hunting export for agent 003, rule 5710

Appendix B — Reference

- Wazuh Documentation: <https://documentation.wazuh.com>
- MITRE ATT&CK: <https://attack.mitre.org/techniques/T1110/001/>
- Hydra: <https://github.com/vanhauser-thc/thc-hydra>
- PCI DSS Requirement 10: Logging and Monitoring Access to System Components